

Midterm Review: Hypothesis Tests

MATH 213

Group 1

Study 1: Mosquitoes

In a study to investigate the effect of drinking beer on mosquito attraction, 43 men were randomly assigned to drink either a liter of beer or a liter of water, with 25 being assigned to the beer group and 18 to the water group. The number of mosquitoes approaching each man was then recorded, and is available in StatKey (Randomization Hypothesis Test for Difference in Means). We can see that the mean number of mosquitoes for the beer group was 23.6 while for the water group it is 19.222. We wish to see whether these data provide evidence that mosquitoes are attracted to beer drinkers (or, more precisely, that the mean number of mosquitoes approaching a beer drinker is significantly larger than the mean number of mosquitoes approaching a water drinker.)

1. Create some notation for these two sample means, and write down the number for the difference of these means.

xBeer: 23.6 mosquitoes

xWater: 19.222 mosquitos

23.6-19.222= 4.378 difference in sample means.

2. Create related notation for what we're trying to estimate: the difference of population means.

$\mu_{\text{beer}} - \mu_{\text{water}}$

3. State the null and alternate hypotheses for this test, both in words and in the correct notation. Remember the null hypothesis is an equation involving the population parameters, and the alternate is an inequality.

$H_0: \mu_{\text{beer}} - \mu_{\text{water}} = 0$

The beverage drank has no effect on mosquito attraction

$H_A: \mu_{\text{beer}} - \mu_{\text{water}} > 0$

The people who drank beer had more mosquitoes approach than people who drank water

4. Use [StatKey](#) to create a randomization distribution for this test by going to the “Test for Difference in Means” link in the hypothesis test section. Notice that the bell-shaped dotplot is centered around the value 0, which represented the null hypothesis. Use the randomization distribution to indicate whether each of the following possible differences in means is (i) very likely to occur just by random chance, (ii) relatively unlikely to occur but might occur occasionally, or (iii) very unlikely to ever occur just by random chance:

–7 (very unlikely)

1 (highly likely)

–3 (not likely)

–0.5 (Very likely)

5 (very unlikely)

5. Locate the actual difference in means observed in the study and shown for the Original Sample. Where does the sample statistic lie in the randomization distribution? Is it likely or unlikely to occur just by random chance?

It is very unlikely to occur by random chance.

6. Use the sample statistic to find the p-value: click the “right tail” box, and type your sample statistic in the box that appears underneath. Is the p-value large or small?

The p-value given by our randomization test at the sample difference of 4.378 is 0.

7. Interpret the results in terms of your null and alternate hypotheses. What can you conclude? Why can you conclude that? Write a sentence or two.

Because the p-value for our sample statistic is 0, we can say that the difference of mean mosquito attraction between the Beer and Water groups that were found in our sample was very unlikely to have been caused by random chance if the null hypothesis is true. Therefore, we have evidence to reject the null hypothesis, and we can conclude that type of beverage may influence mosquito attraction.

Study 2: Lion Attacks

Scientists studying lion attacks on humans in Tanzania found that 95 lion attacks happened between 6 pm and 10 pm within either five days before a full moon or five days after a full moon. Of these, 71 happened during the five days after the full moon while the other 24 happened during the five days before the full moon. Does this sample of lion attacks provide evidence that attacks are more likely after a full moon? In other words, is there evidence that attacks are not equally split between the two five-day periods?

Let p be the proportion of attacks that occur after the full moon. We want to find out if there is evidence that the true value of p is greater than 0.5.

1. What are the null and alternative hypotheses, both in words and with the correct notation?

pBeforeMoon: 24

pAfterMoon: 71

H₀: The proportion of lion attacks that occur after a full moon is .50

$$P_{\text{aftermoon}} = 0.5$$

H_A: The proportion of lion attacks that occur after a full moon is greater than .50

$$P_{\text{aftermoon}} > 0.5$$

2. Use StatKey to do a randomization test and find the p-value. (Note that this is a test for a single proportion since the data come from one sample.)

3. Interpret the results of your hypothesis test: is this evidence that lions really do attack more after a full moon?

Group 2

Study 1: Mosquitoes

In a study to investigate the effect of drinking beer on mosquito attraction, 43 men were randomly assigned to drink either a liter of beer or a liter of water, with 25 being assigned to the beer group and 18 to the water group. The number of mosquitoes approaching each man was then recorded, and is available in StatKey (Randomization Hypothesis Test for Difference in Means). We can see that the mean number of mosquitoes for the beer group was 23.6 while for the water group it is 19.222. We wish to see whether these data provide evidence that mosquitoes are attracted to beer drinkers (or, more precisely, that the mean number of mosquitoes approaching a beer drinker is significantly larger than the mean number of mosquitoes approaching a water drinker.)

1. Create some notation for these two sample means, and write down the number for the difference of these means.

Bear group mean = 23.6

Water group mean = 19.222

Difference in mean ($\bar{x}$) = 4.378

2. Create related notation for what we're trying to estimate: the difference of population means.

μ (μ)

3. State the null and alternate hypotheses for this test, both in words and in the correct notation. Remember the null hypothesis is an equation involving the population parameters, and the alternate is an inequality.

The null hypothesis is that the mosquitoes randomly approached either the beer or water groups (there is no difference between the groups).

$$H_0: \mu_{\text{beer}} - \mu_{\text{water}} = 0$$

The alternate hypothesis is that the mosquitoes are more attracted to the beer group.

$$H_A: \mu_{\text{beer}} - \mu_{\text{water}} > 0$$

4. Use [StatKey](#) to create a randomization distribution for this test by going to the “Test for Difference in Means” link in the hypothesis test section. Notice that the bell-shaped dotplot is centered around the value 0, which represented the null hypothesis. Use the randomization distribution to indicate whether each of the following possible differences in means is (i) very likely to occur just by random chance, (ii) relatively unlikely to occur but might occur occasionally, or (iii) very unlikely to ever occur just by random chance:

-7	1	-3	-0.5	5
Very unlikely (~0%)	Pretty likely (~25%)	Relatively unlikely (~2%)	Very likely (~30%)	Very unlikely (~0%)

5. Locate the actual difference in means observed in the study and shown for the Original Sample. Where does the sample statistic lie in the randomization distribution? Is it likely or unlikely to occur just by random chance?

It is on the very right end of the randomization distribution. It looks unlikely to occur just by random chance.

6. Use the sample statistic to find the p-value: click the “right tail” box, and type your sample statistic in the box that appears underneath. Is the p-value large or small?

The p-value for the original sample is 0.001. This value is small, which means the observed sample data is very unlikely to occur just by random chance (only a 0.1% chance).

7. Interpret the results in terms of your null and alternate hypotheses. What can you conclude? Why can you conclude that? Write a sentence or two.

We can conclude the the observed sample data is very unlikely to occur just by random chance, which means that we can reject the null hypothesis.

Study 2: Lion Attacks

Scientists studying lion attacks on humans in Tanzania found that 95 lion attacks happened between 6 pm and 10 pm within either five days before a full moon or five days after a full moon. Of these, 71 happened during the five days after the full moon while the other 24 happened during the five days before the full moon. Does this sample of lion attacks provide evidence that attacks are more likely after a full moon? In other words, is there evidence that attacks are not equally split between the two five-day periods?

Let p be the proportion of attacks that occur after the full moon. We want to find out if there is evidence that the true value of p is greater than 0.5.

1. What are the null and alternative hypotheses, both in words and with the correct notation?

The null hypothesis is that the lions are equally likely to attack humans before a full moon and after a full moon.

$$H_0: p_{\text{after}} = 0.5$$

The alternate hypothesis is that the lions are more likely to attack humans after a full moon.

$$H_A: p_{\text{after}} > 0.5$$

2. Use StatKey to do a randomization test and find the p-value. (Note that this is a test for a single proportion since the data come from one sample.)

The proportion of lions who attacked after a full moon is 0.747 (74.7%). The p-value is about 0.

3. Interpret the results of your hypothesis test: is this evidence that lions really do attack more after a full moon?

According to the randomization distribution, the original sample data is very unlikely to be observed under the assumption of the null hypothesis (that lions are equally likely to attack humans before and after a full moon). Therefore, the null hypothesis can be rejected, but it may still be difficult to consider it evidence that lions really attack more after a full moon.

Group 3

Study 1: Mosquitoes

In a study to investigate the effect of drinking beer on mosquito attraction, 43 men were randomly assigned to drink either a liter of beer or a liter of water, with 25 being assigned to the beer group and 18 to the water group. The number of mosquitoes approaching each man was then recorded, and is available in StatKey (Randomization Hypothesis Test for Difference in Means). We can see that the mean number of mosquitoes for the beer group was 23.6 while for the water group it is 19.222. We wish to see whether these data provide evidence that mosquitoes are attracted to beer drinkers (or, more precisely, that the mean number of mosquitoes approaching a beer drinker is significantly larger than the mean number of mosquitoes approaching a water drinker.)

1. Create some notation for these two sample means, and write down the number for the difference of these means.
 $X_{\text{beer}} = 23.6$
 $X_{\text{water}} = 19.222$
The observed difference between the two are $23.6 - 19.222 = 4.738$. So the difference of sample means in 4.738 mosquitos.
2. Create related notation for what we're trying to estimate: the difference of population means.
 $\mu_{\text{beer}} - \mu_{\text{water}}$
 μ_{beer} = the true number of mosquitos attracted to beer drinkers
 μ_{water} = the true number of mosquitos attracted to water drinkers

3. State the null and alternate hypotheses for this test, both in words and in the correct notation.

Remember the null hypothesis is an equation involving the population parameters, and the alternate is an inequality.

Null hypothesis (H_0) - There is no difference in the mean number of mosquitos attracted to beer drinkers and water drinkers.

$$H_0: \mu_{\text{beer}} - \mu_{\text{water}} = 0$$

Alternate hypothesis (H_a) - Mosquitoes are more attracted to beer drinkers compared to water drinkers.

$$H_a: \mu_{\text{beer}} - \mu_{\text{water}} > 0$$

4. Use [StatKey](#) to create a randomization distribution for this test by going to the “Test for Difference in Means” link in the hypothesis test section. Notice that the bell-shaped dotplot is centered around the value 0, which represented the null hypothesis. Use the randomization distribution to indicate whether each of the following possible differences in means is (i) very likely to occur just by random chance, (ii) relatively unlikely to occur but might occur occasionally, or (iii) very unlikely to ever occur just by random chance:

−7 - iii

1 - i

−3 - ii

−0.5 - i

5- iii

5. Locate the actual difference in means observed in the study and shown for the Original Sample. Where does the sample statistic lie in the randomization distribution? Is it likely or unlikely to occur just by random chance?
It occurs in the very unlikely to ever occur by random chance.

6. What does each dot in the dotplot represent.

Each plot represents the difference of means in a simulated sample assuming the null hypothesis is true.

6. Use the sample statistic to find the p-value: click the “right tail” box, and type your sample statistic in the box that appears underneath. Is the p-value large or small?

The p-value is very small.

7. Interpret the results in terms of your null and alternate hypotheses. What can you conclude? Why can you conclude that? Write a sentence or two.

Since the p-value is basically 0%, we have strong evidence to reject the null hypothesis. This suggests that beer drinking attracts significantly more mosquitoes on average than water drinkers. The observed difference was unlikely to ever occur due to random chance.

Study 2: Lion Attacks

Scientists studying lion attacks on humans in Tanzania found that 95 lion attacks happened between 6 pm and 10 pm within either five days before a full moon or five days after a full moon. Of these, 71 happened during the five days after the full moon while the other 24 happened during the five days before the full moon. Does this sample of lion attacks provide evidence that attacks are more likely after a full moon? In other words, is there evidence that attacks are not equally split between the two five-day periods?

Let p be the proportion of attacks that occur after the full moon. We want to find out if there is evidence that the true value of p is greater than 0.5.

1. What are the null and alternative hypotheses, both in words and with the correct notation?

Null hypothesis (H_0) - The proportion of lion attacks after the full moon is equal to the proportion before the full moon. In other words, lions are equally likely to attack in the five days before or after a full moon.

$$H_0: p = 0.5$$

Alternate hypothesis (H_a) - The proportion of lion attacks after the full moon is greater than the proportion before the full moon. That is, lions are more likely to attack in the five days after a full moon.

$$H_a: p > 0.5$$

2. Use StatKey to do a randomization test and find the p-value. (Note that this is a test for a single proportion since the data come from one sample.)

3. Interpret the results of your hypothesis test: is this evidence that lions really do attack more after a full moon?

The extremely small p-value means that a sample statistic as large as ours is very unlikely to occur due to random chance if the null hypothesis is true. So this experiment is evidence to reject the null hypothesis, and conclude that 5 days after a full moon does affect lion attack rates.

Group 4

Study 1: Mosquitoes

In a study to investigate the effect of drinking beer on mosquito attraction, 43 men were randomly assigned to drink either a liter of beer or a liter of water, with 25 being assigned to the beer group and 18 to the water group. The number of mosquitoes approaching each man was then recorded, and is available in StatKey (Randomization Hypothesis Test for Difference in Means). We can see that the mean number of mosquitoes for the beer group was 23.6 while for the water group it is 19.222. We wish to see whether these data provide evidence that mosquitoes are attracted to beer drinkers (or, more precisely, that the mean number of mosquitoes approaching a beer drinker is significantly larger than the mean number of mosquitoes approaching a water drinker.)

1. Create some notation for these two sample means, and write down the number for the difference of these means.
 - a. $\bar{X}_{\text{beer}} - \bar{X}_{\text{water}} = 23.6 - 19.2 = 4.4$
2. Create related notation for what we're trying to estimate: the difference of population means.
 - a. $\mu_{\text{beer}} - \mu_{\text{water}} = ?$

3. State the null and alternate hypotheses for this test, both in words and in the correct notation. Remember the null hypothesis is an equation involving the population parameters, and the alternate is an inequality.

Null: There is no difference in the amount of mosquitos that approach a water-drinker compared to a beer-drinker.

$$\mu_{\text{beer}} - \mu_{\text{water}} = 0$$

Alternate: The mean amount of mosquitos that bit the individuals in their respective groups is directly affected by their beverage of choice.

$$\mu_{\text{beer}} - \mu_{\text{water}} > 0$$

4. Use [StatKey](#) to create a randomization distribution for this test by going to the “Test for Difference in Means” link in the hypothesis test section. Notice that the bell-shaped dotplot is centered around the value 0, which represented the null hypothesis. Use the randomization distribution to indicate whether each of the following possible differences in means is (i) very likely to occur just by random chance, (ii) relatively unlikely to occur but might occur occasionally, or (iii) very unlikely to ever occur just by random chance:

−7(iii)

1(i)

−3(ii)

−0.5(i)

5(iii)

5. Locate the actual difference in means observed in the study and shown for the Original Sample. Where does the sample statistic lie in the randomization distribution? Is it likely or unlikely to occur just by random chance?

A difference in means as extreme as 4.4 is extremely unlikely to occur just by random chance. With 2000 simulations it occurred only once giving a less than 1% chance of occurring by chance under the assumption that the null hypothesis is true.

6. Use the sample statistic to find the p-value: click the “right tail” box, and type your sample statistic in the box that appears underneath. Is the p-value large or small?

P-value: .0005 (very small)

7. Interpret the results in terms of your null and alternate hypotheses. What can you conclude? Why can you conclude that? Write a sentence or two.

We can reject the null hypothesis as being true because the odds of our experimental result occurring due to chance under the assumption of the null is >1% out of 2000 simulations making this extremely unlikely. Further studies are warranted to confirm the alternate hypothesis.

Study 2: Lion Attacks

Scientists studying lion attacks on humans in Tanzania found that 95 lion attacks happened between 6 pm and 10 pm within either five days before a full moon or five days after a full moon. Of these, 71 happened during the five days after the full moon while the other 24 happened during the five days before the full moon. Does this sample of lion attacks provide evidence that attacks are more likely after a full moon? In other words, is there evidence that attacks are not equally split between the two five-day periods?

Let p be the proportion of attacks that occur after the full moon. We want to find out if there is evidence that the true value of p is greater than 0.5.

1. What are the null and alternative hypotheses, both in words and with the correct notation?

Null: Lion attacks will happen with no difference resulting from the time relative to the number of days before or after the full moon.

$$H_0: P_{(-5)} - P_{(+5)} = 0$$

Alternate: More lion attacks occur after the full moon

$$H_a: p_{\text{after}} - p_{\text{before}} > 0$$

2. Use StatKey to do a randomization test and find the p-value. (Note that this is a test for a single proportion since the data come from one sample.)

3. Interpret the results of your hypothesis test: is this evidence that lions really do attack more after a full moon?

Group 5

In a study to investigate the effect of drinking beer on mosquito attraction, 43 men were randomly assigned to drink either a liter of beer or a liter of water, with 25 being assigned to the beer group and 18 to the water group. The number of mosquitoes approaching each man was then recorded, and is available in StatKey (Randomization Hypothesis Test for Difference in Means). We can see that the mean number of mosquitoes for the beer group was 23.6 while for the water group it is 19.222. We wish to see whether these data provide evidence that mosquitoes are attracted to beer drinkers (or, more precisely, that the mean number of mosquitoes approaching a beer drinker is significantly larger than the mean number of mosquitoes approaching a water drinker.)

1. Create some notation for these two sample means, and write down the number for the difference of these means.

$$\bar{x}_{\text{beer}} - \bar{x}_{\text{water}} = 4.378$$

$$4.378$$

2. Create related notation for what we're trying to estimate: the difference of population means.
3. $\mu_{\text{beer}} - \mu_{\text{water}} = 0$

3. State the null and alternate hypotheses for this test, both in words and in the correct notation. Remember the null hypothesis is an equation involving the population parameters, and the alternate is an inequality.

$$\mu_{\text{beer}} - \mu_{\text{water}} = 0 \text{ null:}$$

The beer has no effect on the mosquitoes and they are not more attracted to it

$$\mu_{\text{beer}} - \mu_{\text{water}} > 0 \text{ alternative:}$$

The beer does have an effect on the mosquitoes and it attracts them much more.

4. Use [StatKey](#) to create a randomization distribution for this test by going to the “Test for Difference in Means” link in the hypothesis test section. Notice that the bell-shaped dotplot is centered around the value 0, which represented the null hypothesis. Use the randomization distribution to indicate whether each of the following possible differences in means is (i) very likely to occur just by random chance, (ii) relatively unlikely to occur but might occur occasionally, or (iii) very unlikely to ever occur just by random chance:

-7	1	-3	-0.5	5
iii	i	ii	i	ii

5. Locate the actual difference in means observed in the study and shown for the Original Sample. Where does the sample statistic lie in the randomization distribution? Is it likely or unlikely to occur just by random chance?

Its more unlikely to occur, there were very few points near the difference in means.

5a)each dot in the dotplot represent the difference in means from every sample taken randomly based on if mosquitos are attracted to beer.

6. Use the sample statistic to find the p-value: click the “right tail” box, and type your sample statistic in the box that appears underneath. Is the p-value large or small?

The p-value is very small, it is .00033

7. Interpret the results in terms of your null and alternate hypotheses. What can you conclude? Why can you conclude that? Write a sentence or two.

The extremely small p-value means that a sample statistic as large as ours is extremely unlikely to occur due to random chance if the null hypothesis is true. The number was .00033 and this gives us evidence that we can reject the null hypothesis.

Study 2: Lion Attacks

Scientists studying lion attacks on humans in Tanzania found that 95 lion attacks happened between 6 pm and 10 pm within either five days before a full moon or five days after a full moon. Of these, 71 happened during the five days after the full moon while the other 24 happened during the five days before the full moon. Does this sample of lion attacks provide evidence that attacks are more likely after a full moon? In other words, is there evidence that attacks are not equally split between the two five-day periods?

Let p be the proportion of attacks that occur after the full moon. We want to find out if there is evidence that the true value of p is greater than 0.5.

1. What are the null and alternative hypotheses, both in words and with the correct notation?

$H_0: p = 0.5$ $H_a: p > 0.5$

2. Use StatKey to do a randomization test and find the p-value. (Note that this is a test for a single proportion since the data come from one sample.)

3. Interpret the results of your hypothesis test: is this evidence that lions really do attack more after a full moon?

Group 6

Study 1: Mosquitoes

In a study to investigate the effect of drinking beer on mosquito attraction, 43 men were randomly assigned to drink either a liter of beer or a liter of water, with 25 being assigned to the beer group and 18 to the water group. The number of mosquitoes approaching each man was then recorded, and is available in StatKey (Randomization Hypothesis Test for Difference in Means). We can see that the mean number of mosquitoes for the beer group was 23.6 while for the water group it is 19.222. We wish to see whether these data provide evidence that mosquitoes are attracted to beer drinkers (or, more precisely, that the mean number of mosquitoes approaching a beer drinker is significantly larger than the mean number of mosquitoes approaching a water drinker.)

1. Create some notation for these two sample means, and write down the number for the difference of these means.

$$\bar{x}_{\text{beer}} - \bar{x}_{\text{water}} = 4.378$$

2. Create related notation for what we're trying to estimate: the difference of population means.

$$\mu_{\text{beer}} - \mu_{\text{water}} = 0$$

3. State the null and alternate hypotheses for this test, both in words and in the correct notation. Remember the null hypothesis is an equation involving the population parameters, and the alternate is an inequality.

$H_0 = \mu_{\text{beer}} - \mu_{\text{water}} = 0$ Beverage has no effect on the mosquitoes behavior

$H_a = \mu_{\text{beer}} - \mu_{\text{water}} > 0$ Beverage has a effect on the mosquitoes behavior

4. Use [StatKey](#) to create a randomization distribution for this test by going to the “Test for Difference in Means” link in the hypothesis test section. Notice that the bell-shaped dotplot is centered around the value 0, which represented the null hypothesis. Use the randomization distribution to indicate whether each of the following possible differences in means is (i) very likely to occur just by random chance, (ii) relatively unlikely to occur but might occur occasionally, or (iii) very unlikely to ever occur just by random chance:

-7	1	-3	-0.5	5
Very unlikely	likely	relatively unlikely	very likely	Very unlikely

5. Locate the actual difference in means observed in the study and shown for the Original Sample. Where does the sample statistic lie in the randomization distribution? Is it likely or unlikely to occur just by random chance?

The sample statistic that we have observed does show up on the graph but in relation to the data it is very unlikely occur so we can reject the null hypothesis

6. Use the sample statistic to find the p-value: click the “right tail” box, and type your sample statistic in the box that appears underneath. Is the p-value large or small?

The p-value is very small

7. Interpret the results in terms of your null and alternate hypotheses. What can you conclude? Why can you conclude that? Write a sentence or two.

We conclude in the graph that we can reject the null hypothesis. We can conclude this because our sample statistic is very extreme on the graph at 4.378. The p-value that we have found is very small at 0.00067 which represents that our sample statistic is very unlikely to happen.

Study 2: Lion Attacks

Scientists studying lion attacks on humans in Tanzania found that 95 lion attacks happened between 6 pm and 10 pm within either five days before a full moon or five days after a full moon. Of these, 71 happened during the five days after the full moon while the other 24 happened during the five days before the full moon. Does this sample of lion attacks provide evidence that attacks are more likely after a full moon? In other words, is there evidence that attacks are not equally split between the two five-day periods?

Let p be the proportion of attacks that occur after the full moon. We want to find out if there is evidence that the true value of p is greater than 0.5.

1. What are the null and alternative hypotheses, both in words and with the correct notation?

$H_0 = p_{\text{after moon}} - p_{\text{before moon}} = 0$ There is no difference between after the full moon or before the full moon

$H_a = p_{\text{after moon}} - p_{\text{before moon}} > 0$ There is a difference between whether it is before a full moon or after

2. Use StatKey to do a randomization test and find the p-value. (Note that this is a test for a single proportion since the data come from one sample.)

3. Interpret the results of your hypothesis test: is this evidence that lions really do attack more after a full moon?

Group 7

Study 1: Mosquitoes

In a study to investigate the effect of drinking beer on mosquito attraction, 43 men were randomly assigned to drink either a liter of beer or a liter of water, with 25 being assigned to the beer group and 18 to the water group. The number of mosquitoes approaching each man was then recorded, and is available in StatKey (Randomization Hypothesis Test for Difference in Means). We can see that the mean number of mosquitoes for the beer group was 23.6 while for the water group it is 19.222. We wish to see whether these data provide evidence that mosquitoes are attracted to beer drinkers (or, more precisely, that the mean number of mosquitoes approaching a beer drinker is significantly larger than the mean number of mosquitoes approaching a water drinker.)

1. Create some notation for these two sample means, and write down the number for the difference of these means.

$U(\text{beer})$

$U(\text{water})$

Difference of means: 4.378

2. Create related notation for what we're trying to estimate: the difference of population means.

$U(\text{beer}) - U(\text{water})$

3. State the null and alternate hypotheses for this test, both in words and in the correct notation. Remember the null hypothesis is an equation involving the population parameters, and the alternate is an inequality.

Word:

H0: Drinking beer has no effect on attraction of mosquitoes

H1: drinking beer has an effect on attraction of mosquitoes

Notation:

H0: $U = 0$

H1: $U > 0$

4. Use [StatKey](#) to create a randomization distribution for this test by going to the “Test for Difference in Means” link in the hypothesis test section. Notice that the bell-shaped dotplot is centered around the value 0, which represented the null hypothesis. Use the randomization distribution to indicate whether each of the following possible differences in means is (i) very likely to occur just by random chance, (ii) relatively unlikely to occur but might occur occasionally, or (iii) very unlikely to ever occur just by random chance:

-7 $p = 0$	1 $p = 0.248$	-3 $p = 0.01$	-0.5 $p = 0.388$	5 $p = 0$
Impossible	relatively unlikely	Highly unlikely	Almost likely	Impossible

5. Locate the actual difference in means observed in the study and shown for the Original Sample. Where does the sample statistic lie in the randomization distribution? Is it likely or unlikely to occur just by random chance?

$P = 0.0006$

With a p-value of 0.0006 ($p < 0.05$), it is very unlikely that this will happen just to random chance.

6. Use the sample statistic to find the p-value: click the “right tail” box, and type your sample statistic in the box that appears underneath. Is the p-value large or small?

P = 0.0006. The p value is very small

7. Interpret the results in terms of your null and alternative hypotheses. What can you conclude? Why can you conclude that? Write a sentence or two.

Assuming there is no real difference between drinking water and beer and mosquito attraction, out of 5000 simulations, only 3 had a value more extreme than our calculated mean difference of 4.378, therefore, the chance of randomly getting the results of the study are extremely unlikely.

Study 2: Lion Attacks

Scientists studying lion attacks on humans in Tanzania found that 95 lion attacks happened between 6 pm and 10 pm within either five days before a full moon or five days after a full moon. Of these, 71 happened during the five days after the full moon while the other 24 happened during the five days before the full moon. Does this sample of lion attacks provide evidence that attacks are more likely after a full moon? In other words, is there evidence that attacks are not equally split between the two five-day periods?

Let p be the proportion of attacks that occur after the full moon. We want to find out if there is evidence that the true value of p is greater than 0.5.

1. What are the null and alternative hypotheses, both in words and with the correct notation?

H_0 : Lions attack equally between the 5 days before and after a full moon

H_1 : Lions attack more on the 5 days after a full moon than before

2. Use StatKey to do a randomization test and find the p-value. (Note that this is a test for a single proportion since the data come from one sample.)

71/95 =

$P = 0$, highly unlikely.

3. Interpret the results of your hypothesis test: is this evidence that lions really do attack more after a full moon?

Study 1: Mosquitoes

In a study to investigate the effect of drinking beer on mosquito attraction, 43 men were randomly assigned to drink either a liter of beer or a liter of water, with 25 being assigned to the beer group and 18 to the water group. The number of mosquitoes approaching each man was then recorded, and is available in StatKey (Randomization Hypothesis Test for Difference in Means). We can see that the mean number of mosquitoes for the beer group was 23.6 while for the water group it is 19.222. We wish to see whether these data provide evidence that mosquitoes are attracted to beer drinkers (or, more precisely, that the mean number of mosquitoes approaching a beer drinker is significantly larger than the mean number of mosquitoes approaching a water drinker.)

1. Create some notation for these two sample means, and write down the number for the difference of these means.

$$\bar{x}_{\text{beer}} - \bar{x}_{\text{water}} = 0$$

$$\bar{x}_{\text{beer}} - \bar{x}_{\text{water}} > 0$$

4.378

2. Create related notation for what we're trying to estimate: the difference of population means.

$$\mu_{\text{beer}} - \mu_{\text{water}} = 0$$

$$\mu_{\text{beer}} - \mu_{\text{water}} > 0$$

3. State the null and alternate hypotheses for this test, both in words and in the correct notation. Remember the null hypothesis is an equation involving the population parameters, and the alternate is an inequality.

$\mu_{\text{beer}} - \mu_{\text{water}} = 0$ null:

The beer has no effect on the mosquitoes and they are not more attracted to it

$\mu_{\text{beer}} - \mu_{\text{water}} > 0$ alternative:

The beer does have an effect on the mosquitoes and it attracts them much more.

4. Use [StatKey](#) to create a randomization distribution for this test by going to the “Test for Difference in Means” link in the hypothesis test section. Notice that the bell-shaped dotplot is centered around the value 0, which represented the null hypothesis. Use the randomization distribution to indicate whether each of the following possible differences in means is (i) very likely to occur just by random chance, (ii) relatively unlikely to occur but might occur occasionally, or (iii) very unlikely to ever occur just by random chance:

–7

iii

1

i

–3

ii

–0.5

i

5

iii

5. Locate the actual difference in means observed in the study and shown for the Original Sample. Where does the sample statistic lie in the randomization distribution? Is it likely or unlikely to occur just by random chance?

In the original sample, the difference in means is

6. Use the sample statistic to find the p-value: click the “right tail” box, and type your sample statistic in the box that appears underneath. Is the p-value large or small?

7. Interpret the results in terms of your null and alternate hypotheses. What can you conclude? Why can you conclude that? Write a sentence or two.

Study 2: Lion Attacks

Scientists studying lion attacks on humans in Tanzania found that 95 lion attacks happened between 6 pm and 10 pm within either five days before a full moon or five days after a full moon. Of these, 71 happened during the five days after the full moon while the other 24 happened during the five days before the full moon. Does this sample of lion attacks provide evidence that attacks are more likely after a full moon? In other words, is there evidence that attacks are not equally split between the two five-day periods?

Let p be the proportion of attacks that occur after the full moon. We want to find out if there is evidence that the true value of p is greater than 0.5.

1. What are the null and alternative hypotheses, both in words and with the correct notation?
2. Use StatKey to do a randomization test and find the p-value. (Note that this is a test for a single proportion since the data come from one sample.)

3. Interpret the results of your hypothesis test: is this evidence that lions really do attack more after a full moon?

Group 9

Study 1: Mosquitoes

In a study to investigate the effect of drinking beer on mosquito attraction, 43 men were randomly assigned to drink either a liter of beer or a liter of water, with 25 being assigned to the beer group and 18 to the water group. The number of mosquitoes approaching each man was then recorded, and is available in StatKey (Randomization Hypothesis Test for Difference in Means). We can see that the mean number of mosquitoes for the beer group was 23.6 while for the water group it is 19.222. We wish to see whether these data provide evidence that mosquitoes are attracted to beer drinkers (or, more precisely, that the mean number of mosquitoes approaching a beer drinker is significantly larger than the mean number of mosquitoes approaching a water drinker.)

1. Create some notation for these two sample means, and write down the number for the difference of these means.

$x_{\text{beer}} = 23.6$

$x_{\text{water}} = 19.222$

$23.6 - 19.222 = 4.378$ = the difference between the sample means

Create related notation for what we're trying to estimate: the difference of population means.

3. State the null and alternate hypotheses for this test, both in words and in the correct notation. Remember the null hypothesis is an equation involving the population parameters, and the alternate is an inequality.

Null: $\mu_{\text{water}} - \mu_{\text{beer}} = 0$ the liquid consumed has no effect on the attraction of misquitos

Alt: $\mu_{\text{water}} - \mu_{\text{beer}} > 0$ the liquid consumed did have an effect on the mosquitoes attracted: beer was more likely to attract them.

4. Use [StatKey](#) to create a randomization distribution for this test by going to the “Test for Difference in Means” link in the hypothesis test section. Notice that the bell-shaped dotplot is centered around the value 0, which represented the null hypothesis. Use the randomization distribution to indicate whether each of the following possible differences in means is (i) very likely to occur just by random chance, (ii) relatively unlikely to occur but might occur occasionally, or (iii) very unlikely to ever occur just by random chance:

–7:3

1 :1

–3:2

–0.5:1

5:3

5. Locate the actual difference in means observed in the study and shown for the Original Sample. Where does the sample statistic lie in the randomization distribution? Is it likely or unlikely to occur just by random chance?

Very unlikely as it lands on the very far end of the right tail of the data.

6. Use the sample statistic to find the p-value: click the “right tail” box, and type your sample statistic in the box that appears underneath. Is the p-value large or small?

Very small

7. Interpret the results in terms of your null and alternate hypotheses. What can you conclude? Why can you conclude that? Write a sentence or two.

Given that when we assume that the null hypothesis is true, the graph is only able to produce one singular point out of a thousand that could even hope to be considered as greater than or equal to our discovered difference of 4.378, it's very unlikely that this point could be gathered at random. As such, it's likely that this test rejects the null hypothesis

Study 2: Lion Attacks

Scientists studying lion attacks on humans in Tanzania found that 95 lion attacks happened between 6 pm and 10 pm within either five days before a full moon or five days after a full moon. Of these, 71 happened during the five days after the full moon while the other 24 happened during the five days before the full moon. Does this sample of lion attacks provide evidence that attacks are more likely after a full moon? In other words, is there evidence that attacks are not equally split between the two five-day periods?

Let p be the proportion of attacks that occur after the full moon. We want to find out if there is evidence that the true value of p is greater than 0.5.

1. What are the null and alternative hypotheses, both in words and with the correct notation?
null= $p = 0.5$ the time before or after a full moon has no affect on the rate of attacks by lions

Alt: $p > 0.5$ the time before or after a full moon has an affect on the rate of lion attacks: lions are more likely to attack after a full moon.

2. Use StatKey to do a randomization test and find the p-value. (Note that this is a test for a single proportion since the data come from one sample.)

3. Interpret the results of your hypothesis test: is this evidence that lions really do attack more after a full moon?

According to the provided data, the attacks after a full moon is 71 out of 95, meaning our observed proportion is .747. If we were to create a graph assuming the null hypothesis is true and provide 1000 random samples, then it seems as though there is no point that hopes to be greater than or equal to that observed point. As such, it's highly unlikely that such a proportion could have been produced by mere randomness. As a result, it's very likely that this circumstance rejects the null hypothesis.

Group 10

Study 1: Mosquitoes

In a study to investigate the effect of drinking beer on mosquito attraction, 43 men were randomly assigned to drink either a liter of beer or a liter of water, with 25 being assigned to the beer group and 18 to the water group. The number of mosquitoes approaching each man was then recorded, and is available in StatKey (Randomization Hypothesis Test for Difference in Means). We can see that the mean number of mosquitoes for the beer group was 23.6 while for the water group it is 19.222. We wish to see whether these data provide evidence that mosquitoes are attracted to beer drinkers (or, more precisely, that the mean number of mosquitoes approaching a beer drinker is significantly larger than the mean number of mosquitoes approaching a water drinker.)

1. Create some notation for these two sample means, and write down the number for the difference of these means.

x_{beer} : 23.6 bites

x_{water} : 19.222 bites

difference= 4.378 bites

2. Create related notation for what we're trying to estimate: the difference of population means.

3. State the null and alternate hypotheses for this test, both in words and in the correct notation. Remember the null hypothesis is an equation involving the population parameters, and the alternate is an inequality.

H_0 : the mosquitoes randomly attacked the men who drink beer more than the men who didn't drink beer

H_a : the beer that the men drank had a direct correlation with the number of mosquito bites that they received

4. Use [StatKey](#) to create a randomization distribution for this test by going to the “Test for Difference in Means” link in the hypothesis test section. Notice that the bell-shaped dotplot is centered around the value 0, which represented the null hypothesis. Use the randomization distribution to indicate whether each of the following possible differences in means is (i) very likely to occur just by random chance, (ii) relatively unlikely to occur but might occur occasionally, or (iii) very unlikely to ever occur just by random chance:

–7 iii

1 i

–3 ii

–0.5 i

5 iii

5. Locate the actual difference in means observed in the study and shown for the Original Sample. Where does the sample statistic lie in the randomization distribution? Is it likely or unlikely to occur just by random chance?

The proportion of random samples that have a difference in means of 4.378 is 0.001 this means that the chance that the men who drank the beer were randomly selected by the mosquitoes is 0.1%. This means that it is extremely unlikely that the null hypothesis is true.

6. Use the sample statistic to find the p-value: click the “right tail” box, and type your sample statistic in the box that appears underneath. Is the p-value large or small?

The p value is small

7. Interpret the results in terms of your null and alternate hypotheses. What can you conclude? Why can you conclude that? Write a sentence or two.

Given this information we can conclude that the alternate hypothesis is most likely true. We can conclude this because the chance that the mosquitoes are consistently selecting to bite the people that drank the beer first and the chance that they randomly did that in our data is 0.1%

Study 2: Lion Attacks

Scientists studying lion attacks on humans in Tanzania found that 95 lion attacks happened between 6 pm and 10 pm within either five days before a full moon or five days after a full moon. Of these, 71 happened during the five days after the full moon while the other 24 happened during the five days before the full moon. Does this sample of lion attacks provide evidence that attacks are more likely after a full moon? In other words, is there evidence that attacks are not equally split between the two five-day periods?

Let p be the proportion of attacks that occur after the full moon. We want to find out if there is evidence that the true value of p is greater than 0.5.

1. What are the null and alternative hypotheses, both in words and with the correct notation?

H_0 : the lions randomly happen to attack

H_a : the full moon is somehow causing the lions to attack

2. Use StatKey to do a randomization test and find the p-value. (Note that this is a test for a single proportion since the data come from one sample.)

3. Interpret the results of your hypothesis test: is this evidence that lions really do attack more after a full moon?